Literature-Implied Negative Answer to a PGP Variant of Wnuk's Disjoint Test

Run `fa_banach_001`, agent `agent_lane_00`

2026-06-23

Status. `literature_implied_answer`. The proposed positive-Grothendieck analogue of Wnuk's disjoint-test characterization is false. The counterexample is the standard Banach lattice $E = \ell_1$. The relevant separation was already recorded in arXiv:2011.09319, so this packet is classified as a literature-implied answer rather than a new counterexample.

Source question. In Section 1.2 of Alpay–Emelyanov–Gorokhova, *Generalizations of L- and M-weakly compact operators*, arXiv:2206.02718 [1], immediately before Assertion `Wnuk2013`, the authors recall Wnuk's assertion:

$$E \in (\text{DPSP}) \quad \Longleftrightarrow \quad \text{every disjoint } w^*\text{-null sequence in } E'_+ \text{ is norm null.}$$

They ask whether, in this assertion, (DPSP) can be replaced by (PGP) and "norm null" by "weakly null".

Supporting literature. Lourenco–Miranda, *A note on the Banach lattice $c_0(\ell_2^n)$, its dual and its bidual*, arXiv:2011.09319 [2], recall in the introduction that ℓ_1 has the weak Grothendieck property but fails the positive Grothendieck property. The weak Grothendieck property is the condition that every disjoint w^*-null sequence in E' is weakly null, hence it is stronger than the positive-disjoint test in the question.

Direct verification. Let $E = \ell_1$, so $E' = \ell_\infty$ with its usual coordinatewise lattice order, and $(\ell_\infty)' = \text{ba}(\mathbb{N})$.

First, every bounded disjoint sequence in ℓ_∞ is weakly null. Indeed, let $(x_n) \subseteq \ell_\infty$ be bounded and disjoint, say $\|x_n\|_\infty \le M$. Put

$$A_n = \{k \in \mathbb{N} : x_n(k) \neq 0\}.$$

The sets A_n are pairwise disjoint. If $\mu \in \text{ba}(\mathbb{N})$ and $|\mu|$ is its variation, then

$$|\mu(x_n)| \le M\,|\mu|(A_n).$$

Since $|\mu|$ is a finite finitely additive positive measure, pairwise disjointness gives

$$\sum_{j=1}^{N} |\mu|(A_j) \le |\mu|(\mathbb{N}) \quad (N \in \mathbb{N}),$$

and therefore $|\mu|(A_n) \to 0$. Thus $\mu(x_n) \to 0$ for every $\mu \in (\ell_\infty)'$, so $x_n \to 0$ weakly in ℓ_∞. In particular, every disjoint w^*-null sequence in E'_+ is weakly null.

Second, E does not have the positive Grothendieck property. Let

$$u_n = \mathbf{1}_{\{k\in\mathbb{N}:k\geq n\}} \in \ell_{\infty+}.$$

For every $a = (a_k) \in \ell_1$,

$$\langle u_n, a\rangle = \sum_{k\geq n} a_k \to 0,$$

so (u_n) is w^*-null in $E' = \ell_\infty$. However (u_n) is not weakly null. If $\mathcal{U}$ is a free ultrafilter on $\mathbb{N}$, then

$$\Phi_{\mathcal{U}}(x) = \lim_{\mathcal{U}} x_k$$

defines a positive norm-one functional on ℓ_∞, and every cofinite set belongs to $\mathcal{U}$. Hence

$$\Phi_{\mathcal{U}}(u_n) = 1 \quad (n \in \mathbb{N}).$$

So u_n is positive and w^*-null but not weakly null.

Conclusion. The Banach lattice $E = \ell_1$ satisfies the proposed disjoint positive test but fails (PGP). Therefore Wnuk's (DPSP) characterization does not remain true after replacing (DPSP) by (PGP) and norm convergence by weak convergence.

Scope and provenance. This is a full negative answer to the replacement question in arXiv:2206.02718, but it is not recorded as a new run counterexample. The key property separation for ℓ_1 appears in [2]; this packet supplies the direct identification with the 2022 question and a short verification.

References

[1] S. Alpay, E. Emelyanov, and S. Gorokhova, *Generalizations of L- and M-weakly compact operators*, arXiv:2206.02718.

[2] M. L. Lourenco and V. C. C. Miranda, *A note on the Banach lattice $c_0(\ell_2^n)$, its dual and its bidual*, arXiv:2011.09319.